

Arjuna JEE 2025

Maths

DPP: 2

Permutations and Combinations

- Q1** The number of integers greater than 6000 that can be formed, using the digits 3, 5, 6, 7 and 8 without repetition, is
 (A) 120 (B) 72
 (C) 216 (D) 192
- Q2** The value ${}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$, is
 (A) nP_r
 (B) ${}^{n-1}P_r$
 (C) ${}^{n-1}P_{r+1}$
 (D) ${}^nP_{r+1}$
- Q3** If ${}^{l+m}P_2 = 56$, and ${}^{l-m}P_2 = 12$, then $l + m$, is
 (A) 6 (B) 8
 (C) 10 (D) 12
- Q4** The number of ways in which 5 letters be mailed, if there are 3 different mailboxes available, if each letter can be mailed in any mailbox, is
 (A) 81 (B) 243
 (C) 60 (D) 125
- Q5** 3 different railway passes are allotted to 5 students, the number of ways this can be done is
 (A) 60 (B) 20
 (C) 15 (D) 10
- Q6** There are 4 letters and 4 directed envelopes. The number of ways in which all the letters could be put into the wrong envelopes is
 (A) 8 (B) 9
 (C) 16 (D) None of these
- Q7** How many words can be formed out of the letters of the word Article so that the vowels occupy the even places
 (A) 360 (B) 200
 (C) 120 (D) 144
- Q8** The value of $1 + 1 \cdot {}^1P_1 + 2 \cdot {}^2P_2 + 3 \cdot {}^3P_3 + \dots + n \cdot {}^nP_n$, is
 (A) $n!$ (B) $(n-1)!$
 (C) $(n+1)!$ (D) $(n+2)!$
- Q9** A letter lock consists of three rings each marked with 10 different letters. Then, the number of ways in which it is possible to make an unsuccessful attempt to open the lock, is
 (A) 1000
 (B) 999
 (C) 720
 (D) 10000
- Q10** In how many ways can the letters of the word PENCIL be arranged so that
 (i) N is always next to E?
 (ii) N and E are always together?
- Q11** A 5-digit number divisible by 3 is to be formed using the numbers 0, 1, 2, 3, 4 and 5 without repetition. Find total number of ways in which this can be done.
 (A) 196 (B) 216
 (C) 232 (D) 256
- Q12** The number of all five digit numbers which are divisible by 4 that can be formed from the digits 0, 1, 2, 3, 4 (without repetition), is



- (A) 36 (B) 30
(C) 34 (D) None of these

Q13 Find the number of words that can be made out of the letters of the word MOBILE when consonants always occupy odd places.

- (A) 9 (B) 18
(C) 27 (D) 36

Q14 If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800$, then r is

- (A) 39 (B) 41
(C) 28 (D) 43

Q15 Number of ways in which 15 different books can be arranged on a shelf so that two particular books shall not be together is

- (A) $14 \times 15!$
(B) $13 \times 14!$
(C) $14! \times 15!$
(D) $(15!)^2$

Q16 How many words can be made from the letters of the word DELHI, if "L" comes in the middle in every word

- (A) 12 (B) 24
(C) 60 (D) 6

Q17 In how many ways can 5 boys and 3 girls sit in a row so that no two girls are together

- (A) $5! \times 3!$
(B) ${}^4P_3 \times 5!$
(C) ${}^6P_3 \times 5!$
(D) ${}^5P_3 \times 3!$

Q18 Five boys and five girls form a line with the boys and girls alternating. Find the number of ways of making the line.

Q19 In how many ways 5 boys and 3 girls can be seated in a row so that no two girls are together?



Answer Key

Q1 (D)
Q2 (A)
Q3 (B)
Q4 (B)
Q5 (A)
Q6 (B)
Q7 (D)
Q8 (C)
Q9 (B)
Q10 (i)120

(ii)240

Q11 (B)
Q12 (B)
Q13 (D)
Q14 (B)
Q15 (B)
Q16 (B)
Q17 (C)
Q18 $2(5!)^2$
Q19 14400



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